

The Erdős #249/#257 Lean formalization:

what is proved, what is cited, and what stays open

ai_workflow (upstream of [Plectis](#))

July 2026

Companion to the Lean 4 repository [plectis-lean-erdos249-257](#), pinned at commit [cb82a6c](#).

Abstract

This note describes, in ordinary mathematical language, a machine-checked Lean 4 development around two Erdős irrationality problems: the irrationality of Erdős–Borwein-type series $\sum 1/(b^{a_k} - 1)$ (Erdős #257), and the still-open question of whether the totient constant $S = \sum_{n \geq 1} \varphi(n)/2^n$ is irrational (Erdős #249). The development proves the Erdős–Borwein constant irrational for every integer base $b \geq 2$, and the same for several named infinite supports. It rewrites S as a weighted Erdős–Borwein object; proves *unconditionally* that S equals no rational with denominator up to about 7.96×10^{34} ; and *reduces* the irrationality of S to a sequence of finite, decidable certificates. The remaining analytic question is whether that certificate supply is unbounded. Neither open problem is solved. Machine-checked non-claims in the repository state the same boundary. Every formal statement links to the exact Lean source line. The Lean kernel verifies the package with no `sorry`, no `admit`, and no `native_decide`.

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1 Main results and non-results

This section states what the Lean development proves and what it does not. Every item is machine-checked at the pinned commit; later sections expand each one and link it to source. The two named problems are Erdős #249 (is $S = \sum_{n \geq 1} \varphi(n)/2^n$ irrational?) and Erdős #257 (is $\sum_{n \in A} 1/(2^n - 1)$ irrational for *every* infinite $A \subseteq \mathbb{N}$?). Both are open, and this development solves neither.

The Mersenne–Lambert ladder. $L(f) = \sum_{n \geq 1} \frac{f(n)}{2^n - 1}$			
$L(\mu) = \frac{1}{2}$	$L(\varphi) = 2$	$L(1) = E$	$L(\varphi * \mu) = S$ (#249)
rational	rational	irrational (proved)	open
The #249 reduction chain. $S \in \mathbb{Q} \implies$ eventual binary periodicity $\implies R_{N+h} - R_N \in \mathbb{Z}$ $\implies$ a finite certifiedKill witnesses failure $\iff R_{N+h} - R_N \notin \mathbb{Z}$ (completeness: the finite certificate is exact, not a truncation heuristic; #249 becomes “the certificate supply is unbounded”)			

Proved (machine-checked).

- **All-base Erdős–Borwein irrationality.** $\sum_{n \geq 1} 1/(b^n - 1)$ is irrational for every integer $b \geq 2$; the Erdős–Borwein constant E is the base-2 case (Result 4). This is a formalization of Erdős’s classical theorem, and it is exactly the integer-base slice of Erdős #1049.
- **A #257 support calculus.** $\sum_{n \in A} 1/(b^n - 1)$ is irrational, at every base $b \geq 2$, for several structured infinite supports A : factorial, powers of two, multiples, pairwise-coprime with summable reciprocals, eventually periodic, residue classes, and the odd numbers (Result 5). These are named cases, not the universal #257 theorem.
- **A change of coordinates for #249.** The totient constant is a weighted Erdős–Borwein object, $S = L(\varphi * \mu)$ with a nonnegative primitive-conductor weight, and equals $\frac{1}{2} + \sum_d \mu(d)/(2^d - 1)^2$ and $\frac{1}{2} + \Pr(\gcd(X, Y) = 1)$ for independent fair-coin waiting times (Results 2, 3, 7).
- **An unconditional denominator exclusion.** S equals no rational of denominator up to about 7.96×10^{34} (Result 6). This is a record on the denominator, not a proof of irrationality.
- **A reduction of #249 to a finite, decidable supply.** Rationality of S forces an integer tail-period law; a finite certificate witnesses its failure, and by a completeness theorem the certificate exists *exactly* when the tail difference is non-integral (Results 8–11). The initial segment of the required supply is machine-checked non-empty (Result 12).

Not proved, and not claimed. The irrationality of S (#249) and the universal #257 statement are open. The reduction turns #249 into the claim that the certificate supply is *unbounded*. That unboundedness is the remaining analytic question, and it is left open. The repository records these boundaries as machine-checked non-claims (`not_erdos_249_solution`, `not_erdos_257_solution`; [NON_CLAIMS.md](#)).

How it is checked. The Lean 4 kernel verifies the package against Mathlib with no `sorry`, no `admit`, and no `native_decide`; every finite computation is discharged by `decide`. Section 9 gives the reproduction commands.

2 How to read this document

This note is written so that a number theorist who has never used a proof assistant can see what has and has not been proved, and can check any claim without reading Lean. Two conventions make that possible.

First, the mathematics is stated in ordinary notation. Lean names appear only as pointers. Second, every formalized statement carries a link of the form [CertificateKernel.lean:8006](#) (irrational_erdosBorwein_series). The blue `file:line` text links to that line of source on GitHub, pinned to commit [cb82a6c](#); the name in parentheses is the Lean declaration. Line numbers refer to that commit.

The boundary between what is proved, what is only cited, and what is open is stated carefully. The repository enforces it in code through non-claims ([NON_CLAIMS.md](#)); Section 8 repeats it here.

3 Two constants, two problems

Work at base 2 unless a base b is named. Two Dirichlet-type series are central.

- The **Erdős–Borwein constant** $E = \sum_{n \geq 1} \frac{1}{2^n - 1}$.
- The **totient constant** $S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n}$, where φ is Euler’s totient.

The #257 direction. Erdős #257 asks whether $\sum_{n \in A} 1/(2^n - 1)$ is irrational for *every* infinite $A \subseteq \mathbb{N}$. The development proves several cases of the analogous statement $\sum_k 1/(b^{a_k} - 1)$, at every base $b \geq 2$, but not the universal statement.

The #249 direction. Whether S is irrational is open. The development gives an unconditional exclusion of small-denominator rationals and a conditional reduction of the problem to finite certificates. Neither is a solution.

The two constants sit one convolution apart on a single ladder.

4 The Mersenne–Lambert ladder

Define the Lambert-type transform of an arithmetic function f by

$$L(f) = \sum_{n \geq 1} \frac{f(n)}{2^n - 1} = \sum_{m \geq 1} \frac{(f * \mathbf{1})(m)}{2^m}, \quad (f * \mathbf{1})(m) = \sum_{d|m} f(d),$$

the second equality being the standard Lambert rearrangement (one Dirichlet convolution by the constant $\mathbf{1} = \zeta$). Walking the ladder by convolution gives five values with rational, irrational, open, or transcendental status.

input f	$L(f)$	value	status in the repository
μ (Möbius)	$L(\mu)$	$1/2$	rational, proved
φ (totient)	$L(\varphi)$	2	rational, proved
$\mathbf{1}$ (constant one)	$L(\mathbf{1})$	E	irrational, machine-checked
$A = \varphi * \mu$	$L(A)$	S	open (Erdős #249)
Id	$L(\text{Id})$	$\sum \sigma(m)/2^m$	transcendental (Nesterenko 1996 [4], cited only)

The weight $A = \varphi * \mu$ is the primitive-conductor count: $A(p) = p - 2$ on primes, $A \geq 0$, and $A * \zeta = \varphi$. One convolution by ζ separates the open constant S from the rational 2, and one more separates rational from transcendental ($\varphi * \zeta = \text{Id}$, $\text{Id} * \zeta = \sigma$). So S is the rung one convolution from the machine-checked irrational E : the same series $\sum(\cdot)/(2^d - 1)$, with the constant weight 1 replaced by the nonnegative weight A .

Result 1 (the rational rungs). $\sum_{d \geq 1} \mu(d)/(2^d - 1) = 1/2$ and $\sum_{d \geq 1} \varphi(d)/(2^d - 1) = 2$, *exactly*.

Formalized: [MersenneLambertLadder.lean:527](#) (`tsum_moebius_div_two_pow_sub_one_eq_half`) and [MersenneLambertLadder.lean:515](#) (`tsum_totient_div_two_pow_sub_one_eq_two`); restated on the #249 series shape at [tsum_moebius_div_two_pow_sub_one_eq_half](#) and [tsum_totient_div_two_pow_sub_one_eq_two](#).

Result 2 (the positive lift, $L(A) = S$). *With the primitive-conductor weight $A = \varphi * \mu$,*

$$\sum_{d \geq 1} \frac{A(d)}{2^d - 1} = \sum_{n \geq 1} \frac{\varphi(n)}{2^n} = S.$$

Formalized: [CertificateKernel.lean:18116](#) (`tsum_primWeight_div_two_pow_sub_one_eq_totient_series`), with the weight A defined at [primWeight](#). This places S in the same $\sum(\cdot)/(2^d - 1)$ family as E , with a nonnegative weight.

Result 3 (the Möbius-square lens).

$$S = \frac{1}{2} + \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2}.$$

Formalized: [CertificateKernel.lean:18125](#) (`totient_series_eq_half_add_moebius_mersenne_square`). Writing $T = \sum_{d \geq 1} \mu(d)/(2^d - 1)^2$, irrationality of S is equivalent to irrationality of T , since $S = \frac{1}{2} + T$. The factor $1/(2^d - 1)^2$ is the probability that d divides two independent fair-coin waiting times (Section 6.2). The signed constant, the coprime-pair mass, and the tail differences of Section 6.3 are three readings of the same quantity.

5 Erdős–Borwein-type irrationality (the #257 direction)

5.1 Full support: the Erdős–Borwein constant, every base

Result 4 (full-support irrationality). *For every integer $b \geq 2$, the series $\sum_{n \geq 1} \frac{1}{b^n - 1}$ is irrational.*

In particular the Erdős–Borwein constant $E = \sum_{n \geq 1} 1/(2^n - 1)$ is irrational.

Formalized: [CertificateKernel.lean:7999](#) (`irrational_erdosSum_full_support`) (all bases) and [CertificateKernel.lean:8006](#) (`irrational_erdosBorwein_series`) (the base-2 corollary). This is Erdős’s 1948 theorem [1]. In the ladder of Section 4 it is the rung $L(1)$, and it is the fixed machine-checked point one convolution away from the open S .

The proof runs through the certificate machine of the kernel rather than through a direct digit argument: a bounded Bertrand/CRT frame on a first block, a middle-window average over divisor pairs with a pigeonhole selection, and an explicit closure of the parameters. The Lambert bridge $\sum 1/(b^n - 1) = \sum_m \tau(m)/b^m$ (divisor-count series) is [erdosSum_full_support_eq_tsum_divisor_count](#) in the source.

5.2 Named infinite-support cases

Beyond full support, the development proves irrationality for a family of infinite supports A , at every base $b \geq 2$. Each is a formalized case of the #257 statement family. None is the universal statement, and none is claimed as new mathematics.

Result 5 (support-class family). *For every integer $b \geq 2$, the series $\sum_{n \in A} 1/(b^n - 1)$ is irrational for each of the following supports A :*

- (a) *factorial support $A = \{n! : n \geq 1\}$, giving $\sum 1/(b^{n!} - 1)$;*
- (b) *power-of-two support $A = \{2^n : n \geq 0\}$, giving $\sum 1/(b^{2^n} - 1)$;*
- (c) *multiples $A = d\mathbb{N}$;*
- (d) *pairwise-coprime supports with summable reciprocals (Erdős's condition [2]);*
- (e) *eventually-periodic supports, residue classes, and the odd numbers.*

These sort into three groups. The classical case is (d): infinite pairwise-coprime A with $\sum_{a \in A} 1/a < \infty$, formalized from Erdős's condition. The lcm-gap named cases are (a) and (b), whose support gaps outgrow the running lcm; the structured families are (c) and (e).¹ Formalized: (a) `irrational_erdosSum_factorial_support`; (b) `irrational_erdosSum_two_pow_support`; (c) `irrational_erdosSupportSeries_multiples`; (d) `irrational_erdosSupportSeries_pairwise_coprime`; (e) `irrational_erdosSupportSeries_eventuallyPeriodic`, `irrational_erdosSupportSeries_residueClass`, `irrational_erdosSupportSeries_odd`. The two base-2 instances that name the problem directly are `erdos257_family_factorial_instance` and `erdos257_family_two_pow_instance`.

The factorial and power-of-two cases are worth a word because they are not reachable by a Liouville shortcut: the power-of-two gap $2^k - \text{lcm}$ stays within a bounded ratio of the lcm, so the terms do not decay fast enough for a transcendence-measure argument, and a uniform criterion is needed. The mechanism is what the repository calls *period noncollapse*: a prime-valuation-deficit witness forces the multiplicative order of the base modulo the relevant modulus not to collapse, which rules out the long repeated digit blocks a rational value would require.

Remark. A source comment near `lcm_gap_hypothesis_fails_full_support` describes a second, near-integer route to full support whose witnesses are not built there, and says so. The full-support theorem is proved by the separate certificate engine cited under Result 4. The two coexist; the near-integer route is one lane, not a retraction of the theorem.

5.3 What stays open

The universal Erdős #257 statement, irrationality of $\sum_{n \in A} 1/(2^n - 1)$ for *every* infinite A , is not proved. The proved cases are a family, not the quantified theorem, and the repository marks this in code (the non-claim `not_erdos_257_solution` in `NON_CLAIMS.md`).

6 The totient constant S (Erdős #249)

Whether $S = \sum_{n \geq 1} \varphi(n)/2^n$ is irrational is open. The development gives two things and claims neither as a solution: an unconditional exclusion of small-denominator rationals, and a conditional reduction to finite certificates.

¹Periodic-coefficient Lambert series have their own literature; see e.g. Luca–Tachiya [6]. The cases here are formalized in the repository's own framework and are not claimed as new results.

6.1 Unconditional: no small denominator

Result 6 (denominator exclusion). *For every integer p and every q with $1 \leq q \leq 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}$,*

$$S \neq \frac{p}{q}.$$

Equivalently, if S is rational then its reduced denominator exceeds that bound.

Formalized: [CertificateKernel.lean:18055](#) (`tsum_totient_div_pow_two_ne_ratCast_of_den_le_79639646646701375323355774875831053`). The bound is reached by a ladder of finite, elementary exclusions,

$$4838 \rightarrow 4\,194\,304 = 2^{22} \rightarrow 2.49 \times 10^{17} \text{ (Farey } K=120) \rightarrow 7.96 \times 10^{34} \text{ (Farey } K=240),$$

each a mediant/Farey argument that traps any candidate m/q inside a forbidden gap. See [tsu_m_totient_div_pow_two_ne_ratCast_of_den_le_4838](#) for the first rung and [gap_check_window_1_120_le_248672326362367909](#), [gap_check_window_1_240_le_79639646646701375323355774875831053](#) for the two Farey windows.

This is a record on the denominator, not a proof of irrationality. A rational with a larger denominator is not excluded by it.

Remark (one record, four faces). Because the ladder identities of Section 4 are equalities, the same finite record transfers verbatim to three other representations of the constant: to the positive lift $\sum_d A(d)/(2^d - 1)$ ([CertificateKernel.lean:18144](#)), to the signed Möbius-square constant $T = \sum_d \mu(d)/(2^d - 1)^2$ at half the bound ([CertificateKernel.lean:18158](#)), and to the coprimality probability of Section 6.2 at half the bound. Half appears because $S = \frac{1}{2} + T$ turns a denominator d for T into $2d$ for S .

6.2 The geometric picture: S as coprime-pair mass

Let X, Y be independent fair-coin waiting times, $\Pr(X = n) = 2^{-n}$ for $n \geq 1$. Then $\Pr(d \mid X) = 1/(2^d - 1)$, so the Lambert transform is a divisor calculus of X : $L(f) = \mathbb{E}[(f * \zeta)(X)]$. The rungs read as $L(\mu) = \Pr(X = 1) = 1/2$, $L(\varphi) = \mathbb{E}[X] = 2$, $L(\mathbf{1}) = \mathbb{E}[\tau(X)] = E$, and $L(A) = \mathbb{E}[\varphi(X)] = S$.

Counting visible lattice points (a, b) with $a + b = n$, $a > 0$, $\gcd(a, b) = 1$ gives exactly $\varphi(n)$, uniformly in n , so S is itself a coprime-pair mass.

Result 7 (coprimality probability).

$$S = \frac{1}{2} + \Pr(\gcd(X, Y) = 1) = \frac{1}{2} + \sum_{\substack{a, b \geq 1 \\ \gcd(a, b) = 1}} 2^{-(a+b)}.$$

Formalized: [CertificateKernel.lean:18228](#) (`totient_series_eq_half_add_visible_coprime_pairs`), with the underlying lattice identity at [tsum_visible_coprime_pairs_eq_totient_series](#). Base 2 is the one point where $\Pr(X = a) \Pr(Y = b) = 2^{-(a+b)}$ needs no normalizing constant, so the totient sum and the coprimality probability line up with no boundary correction. In this reading, Erdős #249 asks whether two independent fair-coin waiting times are coprime with irrational probability.

6.3 Conditional: reduction to finite certificates

The reduction turns irrationality of S into the non-vanishing of a decidable predicate at infinitely many parameters. Everything in it is elementary: $\varphi(n) \leq n$, geometric tails, and integer window arithmetic. There is no q-*Padé* input, no Stern–Brocot input, and no unformalized citation ([TotientTailPeriodKiller](#)).

Step 1: rationality forces an integer tail-period law. Write the fractional layer of $2^N S$ as the tail

$$R_N = \sum_{m \geq 1} \frac{\varphi(N+m)}{2^m}.$$

If S is rational, its binary expansion is eventually periodic, which forces, for a suitable period h and all large N , the integrality $R_{N+h} - R_N \in \mathbb{Z}$. To refute integrality at a finite depth L , form the window discrepancy

$$\Delta_{h,N,L} = \sum_{j=0}^{L-1} (\varphi(N+h+1+j) - \varphi(N+1+j)) 2^{L-1-j} \in \mathbb{Z},$$

the first L binary digits of $R_{N+h} - R_N$ up to a controlled deep tail (bounded using only $\varphi(n) \leq n$), and call the pair *certified* when the residue of $\Delta_{h,N,L}$ modulo 2^L misses a neighbourhood of 0:

$$\text{certifiedKill}(h, N, L) : \quad N + h + L + 2 < (\Delta_{h,N,L} \bmod 2^L) < 2^L - (N + h + L + 2).$$

This is a finite integer computation, decidable by the kernel. It is the natural finite object here because a rational S would pin the tail difference to an integer, and a residue that stays a fixed distance from 0 is a finite proof that it did not.

Result 8 (tail-period reduction). *If for every period $h \geq 1$ and every threshold N_0 there exist $N \geq N_0$ and L with $\text{certifiedKill}(h, N, L)$, then S is irrational.*

Formalized: [TotientTailPeriodKiller.lean:386](#) (`irrational_totient_series_of_certificate_supply`); the tail R_N and the predicate are [totientTail](#) and [certifiedKill](#).

Step 2: collapse the two parameters to one. Every multiple of a period is a period, so $\text{lcm}(1, \dots, t)$ is the universal-period envelope: any eventual period of a rational S divides it once t is large. Standing on that one diagonal ray covers all possible periods at once, and the position parameter drops out too.

Result 9 (diagonal reduction). *Write $\text{lcm}(1..t)$ for $\text{lcm}(1, \dots, t)$. If for every t_0 there exist $t \geq t_0$ and L with $\text{certifiedKill}(\text{lcm}(1..t), \text{lcm}(1..t), L)$, then S is irrational.*

Formalized: [LcmDiagonalReduction.lean:92](#) (`irrational_totient_series_of_lcm_diagonal_certificate_supply`).

Step 3: the cone, flatness, and completeness. On the cone $\{k \cdot \text{lcm}(1..t)\}$, rationality forces not one pairwise relation but a single fractional constant across the whole cone.

Result 10 (cone flatness). *If S is rational, then the relevant tail values share one fractional part across the whole lcm cone.*

Formalized: [LcmConeFlatness.lean:90](#) (`rational_totient_series_forces_lcm_cone_flatness`).

The certificates are exact, not a lossy heuristic: a certificate exists exactly when the object it certifies is non-integral. That turns the finite computations from experiments into receipts.

Result 11 (certificate completeness). *For all h, N ,*

$$(\exists L, \text{certifiedKill}(h, N, L)) \iff R_{N+h} - R_N \notin \mathbb{Z}.$$

Formalized: [LcmConeFlatness.lean:316](#) (`exists_certifiedKill_iff_tail_diff_notMem_int`). Finite certificates are receipts; non-integrality is the object. A joint refuter can interrogate a menu of cone vertices at once and is sharper than any pairwise test: it refutes the single-fractional-part model that rationality predicts on a finite sample of cone vertices ([LcmConeNonflat.lean:126](#) (`exists_nonintegral_pair_of_coneNonflatCert`)).

The certificates are non-empty. The initial segment of the required supply is machine-checked, so the reduction is not vacuous. By completeness, each of these is a kernel-checked proof of a specific non-integral tail difference, not a numerical experiment.

Result 12 (finite deposits). *The following are checked by the Lean kernel:*

- *certifiedKill*($h, 12, 16$) for every period $h \in [1, 8]$;
- *certified kills* for every period $h \in [1, 16]$ at a fixed window;
- *diagonal certificates* *certifiedKill*($\text{lcm}(1..t), \text{lcm}(1..t), L$) for $t \in [1, 6]$, and at $t = 7, 8$;
- *a tabulated joint cone-vertex certificate.*

Formalized: `certifiedKill_all_small`, `certifiedKill_all_upto_sixteen`, `certifiedKill_periodLcm_diagonal_upto_six` (with `certifiedKill_periodLcm_diagonal_at_seven`, `certifiedKill_periodLcm_diagonal_at_eight`), and `coneNonflatCert_cells`. These deposits show that each layer of the reduction is non-vacuous and, by completeness, exact. They are not evidence that the unbounded supply exists.

What is open. The reduction converts #249 into the statement that the certificate supply is *unbounded*: certificates exist for arbitrarily large parameters. By completeness (Result 11) this is equivalent to the tail differences being non-integral infinitely often, and equivalently to the Farey gap denominators of Section 6.1 being unbounded: the same obstruction seen as a rational denominator, an eventual binary period, and a flat tail on the lcm cone. That unboundedness is the remaining analytic question, and it is left open. The reduction is real; the closure is not asserted.

7 Where this sits: the Erdős landscape

The table records how the development relates to the numbered problems it is near. Statuses are as listed on the individual problem pages erdosproblems.com/{249, 257, 1049, 69, 250, 258} in July 2026; none of the external results is re-proved here.²

problem	status	relation in this repository
#249 $\sum \varphi(n)/2^n$	open	the target; unconditional denominator exclusion and a conditional reduction, no solution
#257 $\sum_{n \in A} 1/(2^n - 1)$	open	named infinite-support cases and the full-support ($A = \mathbb{N}$) case, not the universal statement
#1049 $\sum 1/(t^n - 1)$, $t \in \mathbb{Q}_{>1}$	open (rational t)	the integer-base case $b \geq 2$ is formalized (Result 4); the rational case is untouched
#69 $\sum \omega(n)/2^n$	solved [7]	the prime-support case of #257; only the identity bridge to $\sum_p 1/(2^p - 1)$ is formalized, not the irrationality
#250 $\sum \sigma(n)/2^n$	solved [4]	the ladder neighbour $L(\text{Id})$; cited, not re-proved
#258 $\sum \tau(n)/(a_1 \cdots a_n)$	solved	not addressed here; the monotone φ/σ sequel is the sibling territory of #249

²The irrationality of S (#249) and the universal #257 statement are open. #1049 is open for rational $t > 1$, with the integer-base case attributed to Erdős. #69 (Tao–Teräväinen [7]), #250 (Nesterenko 1996 [4]), and #258 (recorded as proved in Lean) are solved; none is re-proved here.

In short: this is a formalization of Erdős–Borwein-type irrationality together with an unconditional record and a finite-certificate reduction for the open totient constant. It is not a solution to #249 or to the universal #257.

8 What is proved, cited, and open

Proved (machine-checked, no sorry, no admit). Irrationality of $\sum 1/(b^n - 1)$ for every $b \geq 2$ and of the named support cases (Results 4, 5); the ladder identities and the Möbius-square and coprimality reformulations (Results 2, 3, 7); the unconditional denominator exclusion to 7.96×10^{34} (Result 6); the full chain of conditional #249 reductions with certificate completeness and a non-empty initial supply (Results 8–12).

Cited, not formalized here. Transcendence of $\sum \sigma(m)/2^m$ (Nesterenko 1996 [4]) and the q -zeta anchor (Postelmans–Van Assche [5]). They are named as ladder neighbours.

Open, and not solved here. Irrationality of S (#249); the universal #257 statement. The conditional reductions turn #249 into “the certificate supply is unbounded”, which is the remaining analytic question.

The repository states these boundaries in code as machine-checked non-claims: `not_erdos_249_solution`, `not_erdos_257_solution`, and four provenance non-claims ([NON_CLAIMS.md](#)).

9 Verification and reproducibility

The package is checked by the Lean 4 kernel against Mathlib, pinned in [lake-manifest.json](#) at toolchain `leanprover/lean4:v4.29.1`. Three facts matter for trust.

- There is no `sorry` and no `admit` anywhere in the package.
- Every finite computation is discharged by `decide`, checked by the kernel. There is no use of `native_decide`, so no compiler is trusted.
- The package has more than 1,600 theorem declarations. The assembled microkernel is [CertificateKernel.lean](#) (about 18,900 lines); the machine-generated finite certificates are [GeneratedCertificates.lean](#) (about 27,700 lines) plus per-base shards, each instance closed by `decide`.

To build and check:

```
lake exe cache get      # fetch the prebuilt Mathlib cache
lake build              # build and kernel-check the package
```

Trust rests on `lake build` alone: it elaborates every declaration and the Lean 4 kernel checks each proof term, including every `decide`, against Mathlib at the pinned toolchain. There is no separate trusted step and no compiler in the loop. Continuous integration runs the same build on every push.

10 A sibling project, in progress

Separately from this repository, I am working on the [Ramanujan Machine Challenge](#) (ten problems, released July 2026, on proofs or symbolic derivations of explicit formulas for constants such as π , e , Catalan’s constant, and zeta values). That work is a different Lean lane and is not part of this snapshot: no challenge target is claimed solved or formalized here, and nothing in this paper depends on it. A coarse status map for that project lives in the repository README.

A Index of principal declarations

Each row links to the declaration at commit [cb82a6c](#).

statement	source line (and declaration)
$\sum 1/(b^n - 1)$ irrational, all $b \geq 2$	CertificateKernel.lean:7999 irrational_erdosSum_full_support
Erdős–Borwein E irrational	CertificateKernel.lean:8006 irrational_erdosBorwein_series
factorial support	CertificateKernel.lean:5706 irrational_erdosSum_factorial_support
power-of-two support	CertificateKernel.lean:5730 irrational_erdosSum_two_pow_support
pairwise-coprime supports	CertificateKernel.lean:10447 irrational_erdosSupportSeries_pairwise_coprime
$L(\mu) = 1/2$	MersenneLambertLadder.lean:527 tsum_moebius_div_two_pow_sub_one_eq_half
$L(\varphi) = 2$	MersenneLambertLadder.lean:515 tsum_totient_div_two_pow_sub_one_eq_two
positive lift $L(A) = S$	CertificateKernel.lean:18116 tsum_primWeight_div_two_pow_sub_one_eq_totient_series
Möbius-square lens	CertificateKernel.lean:18125 totient_series_eq_half_add_moebius_mersenne_square
$S = \frac{1}{2} + \Pr(\gcd = 1)$	CertificateKernel.lean:18228 totient_series_eq_half_add_visible_coprime_pairs
$S \neq p/q$ for $q \leq 7.96 \times 10^{34}$	CertificateKernel.lean:18055 tsum_totient_div_two_ne_ratCast_of_den_le_79639646646701375323355774875831053
tail-period reduction	TotientTailPeriodKiller.lean:386 irrational_totient_series_of_certificate_supply
diagonal reduction	LcmDiagonalReduction.lean:92 irrational_totient_series_of_lcm_diagonal_certificate_supply
cone flatness from rationality	LcmConeFlatness.lean:90 rational_totient_series_forces_lcm_cone_flatness
certificate completeness	LcmConeFlatness.lean:316 exists_certifiedKill_iff_tail_diff_notMem_int
joint cone refuter	LcmConeNonflat.lean:126 exists_nonintegral_pair_of_coneNonflatCert
non-empty certified-kill segment	TotientTailPeriodKiller.lean:396 certifiedKill_all_small

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